SOLUTION ARCHITECTURE

Date: 15 February 2025
Team ID: SWTID-2026-1751
Project Name: FitHub CRM – Fitness Center Management System

---

1. Introduction

Solution architecture is a structured approach that connects business requirements with technical implementation. It defines how different components of a system interact with each other to deliver the required functionality.

For the FitHub CRM system, the solution architecture provides a clear blueprint that explains how fitness center operations such as membership management, class scheduling, trainer allocation, and booking processes are implemented using Salesforce.

The architecture ensures that the system is scalable, secure, efficient, and easy to maintain. It also helps stakeholders understand how the system works and how data flows across different components.

---

2. Objectives of Solution Architecture

The main objectives of designing the solution architecture are:

- To identify the best technology solution for fitness center management
- To define the structure of the system
- To ensure smooth data flow between components
- To support scalability and performance
- To enable automation of business processes
- To provide a clear understanding to stakeholders

---

3. System Overview

The FitHub CRM system is built on Salesforce and consists of multiple interconnected components:

- User Interface (Lightning App)
- Business Logic (Apex, Flows, Validation Rules)
- Data Layer (Custom Objects and Fields)
- Reporting Layer (Reports and Dashboards)

Each layer plays a specific role in ensuring smooth system operation.

---

4. Architecture Components

4.1 User Interface Layer

This layer is responsible for interaction between users and the system.

- Built using Lightning App Builder
- Provides forms for data entry
- Displays dashboards and reports
- Ensures user-friendly navigation

Users such as members, admins, and managers interact with the system through this layer.

4.2 Application Logic Layer

This layer handles business rules and automation.

Includes:

- Apex Triggers
- Flows
- Validation Rules

Functions:

- Validates booking conditions
- Automates notifications
- Assigns trainers dynamically
- Calculates booking details

This layer ensures that all operations follow business rules.

4.3 Data Layer

This layer stores all system data using custom objects.

Objects:

- Member
- Trainer
- Class
- Booking

Relationships:

- Booking → Member
- Booking → Class
- Class → Trainer

This structured data model ensures efficient storage and retrieval of information.

4.4 Reporting Layer

This layer provides insights and analytics.

- Reports for bookings, members, and revenue
- Dashboards for visualization
- Real-time data updates

Managers use this layer to make decisions.

---

5. Data Flow Architecture

The system follows a structured data flow process:

1. User enters data (member, booking, etc.)
2. Data is processed through validation rules and automation
3. Information is stored in Salesforce database
4. System performs actions (assignment, notification)
5. Output is displayed through UI, reports, and dashboards

---

6. Solution Architecture Diagram

👉 Paste this in Word and convert to SmartArt / shapes for better marks:

```
+----------------------+
|     Users          |
| (Admin / Member)    |
+----------+-----------+
           |
           ↓
+----------------------+
| User Interface      |
| (Lightning App)     |
```

```
+----------+-----------+
        |
        ↓
+----------------------+
| Application Logic    |
| (Flows, Apex, Rules) |
+----------+-----------+
        |
        ↓
+----------------------+
|   Data Layer         |
| (Member, Class, etc) |
+----------+-----------+
        |
        ↓
+----------------------+
| Reports & Dashboard  |
+----------------------+
```

---

7. Features Supported by Architecture

The architecture supports:

- Automated membership tracking
- Class scheduling
- Trainer allocation
- Booking validation
- Payment tracking
- Notification system
- Real-time reporting

---

8. Advantages of Architecture

- Modular design

- High scalability
- Secure data handling
- Efficient automation
- Easy maintenance

---

9. Future Enhancements

The architecture can be extended to include:

- Mobile application integration
- AI-based recommendations
- IoT fitness device integration
- Online payment systems

---

10. Conclusion

The solution architecture of FitHub CRM provides a strong foundation for implementing a scalable and efficient fitness center management system. It ensures smooth interaction between components, supports automation, and delivers a high-quality user experience.